

When hormone patterns change

How can you use the normal feedback system without jumping to a diagnosis?

Learning goal

Explain model changes associated with ovarian ageing, early pregnancy and combined hormonal contraception using hormone sources, targets and feedback.

Before you begin

Learn the normal follicle → ovulation → corpus luteum sequence before using it to predict a change.

How to complete this lesson Learn → worked example → practise → lesson check

- Read the explanation and its diagrams. Open a bold term when you need a definition.
- Follow the worked example, then try both supported questions. Select Check answer for feedback.
- Open the required check and select Start. Correct both selections to unlock writing.
- Write each explanation and select its Save written response button. Editing saved writing makes it a draft again.
- Select Submit and finish to complete the check. Find your records in All My Work.

Expected work: one completed lesson check. Supported practice, videos and textbook questions are optional. Saved writing is not automatically graded.

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Key terms: menopause · hCG · hormonal contraception · negative feedback

Vocabulary help

Key terms

Words for this lesson

Use these definitions when you need help with a term.

menopause

The end of menstrual cycling associated with declining ovarian follicular activity.

hCG

Human chorionic gonadotropin, released by developing trophoblast tissue in early pregnancy.

hormonal contraception

Conception control using hormones or hormone-like substances to alter reproductive processes.

negative feedback

A response that tends to oppose the initial change.

Begin with a pattern you can explain

In the last two lessons, we connected events in the ovary with events in the uterus. Now we will change one part of that system and ask what should follow. Keep three questions separate: What was observed? What mechanism could explain it? What else would we need to know?

The cycle diagram below is our baseline, a simplified cycle without pregnancy. Read the ovarian row from a developing follicle through ovulation to the corpus luteum, the hormone-producing tissue left from the ovulated follicle. In the uterine row, follow the endometrium, the inner lining of the uterus. Growing follicles supply estrogen; after ovulation the corpus luteum supplies progesterone and estrogen. These blood-borne signals help the lining grow and remain ready to support an early embryo.

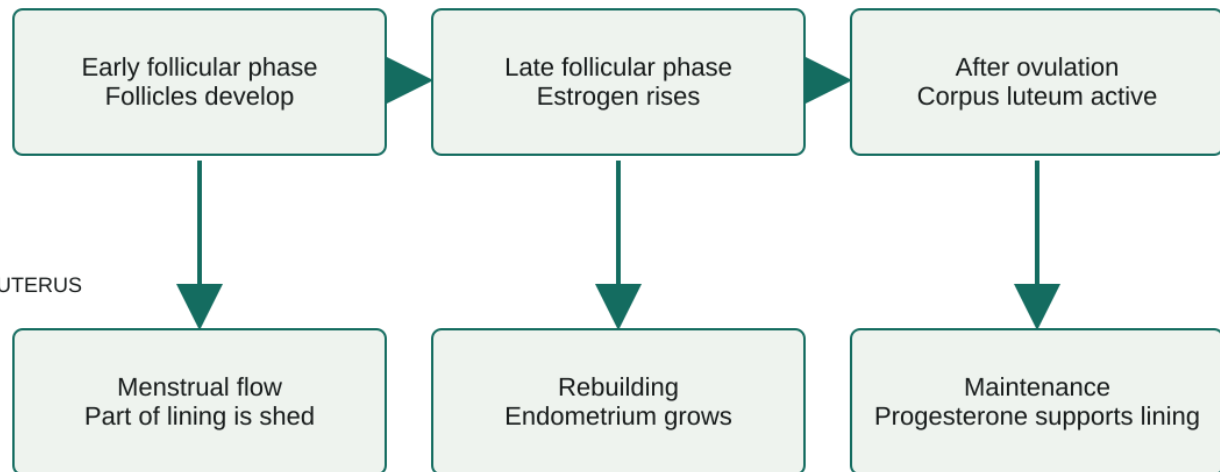
The last column shows an active corpus luteum supporting the lining. Read the footer for the later event that returns this model to flow. Without pregnancy support, the corpus luteum regresses, meaning it shrinks and loses activity. Ovarian hormone output falls, and the functional part of the lining is shed. The first column shows menstruation overlapping new follicular activity; its downward arrow does not mean that a new follicle causes the shedding. The important cause is the preceding fall in hormone support. The picture is a sequence model, not a personal calendar or a picture of every altered cycle.

For example, a low progesterone result means only that the measured sample contained little progesterone. A single sample might have been collected before the usual post-ovulation rise. Even a missing recorded LH peak is not automatically proof that no peak occurred: the brief surge might fall between measurements. With a sufficiently complete pattern, absent surge and absent later progesterone rise can support a hypothesis of disrupted ovulation. They still do not identify one definite cause.

Ovarian and uterine events occur together

Illustrative 28-day model — actual cycle timing varies

OVARY



Flow overlaps early follicular development.

Hormone decline after corpus-luteum regression permits the next flow.

Baseline cycle without pregnancy, reused unchanged from lesson07. Compare the normal regression step with the altered hormone cases in this lesson.

A weaker ovarian response can mean stronger pituitary stimulation

An ovarian follicle contains an immature oocyte and supporting cells. As the supply of functioning follicles declines with ovarian ageing, ovulation becomes less regular. Hormone levels can fluctuate during the transition; they do not simply fall by the same amount every month. Menopause names the end of menstrual cycling associated with the loss of normal ovarian follicular activity. After menopause, ovarian estrogen and progesterone output is much lower than in typical reproductive cycles.

Trace the feedback rather than assuming every hormone must move together. Normally, ovarian hormones contribute inhibitory signals to the hypothalamus and anterior pituitary. Negative feedback means that a response opposes the change that produced it. If ovarian output falls, some of that inhibition is removed. A functioning anterior pituitary can then release more FSH and LH, the gonadotropins that stimulate the gonads. Reduced ovarian inhibin, another inhibitory signal especially important for FSH, also contributes.

So low ovarian output and high pituitary output can occur together. More stimulation does not create a fresh supply of responsive follicles. Picture a signal reaching a target that cannot respond as strongly: increasing the message does not guarantee restoration of the response.

The teacher notes also describe age-related testosterone changes, sometimes called andropause. Do not turn that comparison into an identical male menopause. Testosterone changes are more gradual and variable, and there is no universal sudden end to sperm production. Fatigue or a changed menstrual pattern alone cannot establish a hormone diagnosis. The older textbook treatment discussions are historical reading, not personal treatment guidance.

Early pregnancy keeps the corpus luteum working

Change the baseline at the corpus luteum. After fertilization and early development, the embryo reaches the uterus. Its outer trophoblast tissue participates in implantation, attachment and embedding in the endometrium, and contributes to the placenta. During early pregnancy, trophoblast tissue releases human chorionic gonadotropin, abbreviated hCG.

hCG travels in the blood and supports the corpus luteum. The corpus luteum therefore continues releasing progesterone and estrogen instead of following its usual unsupported decline. Progesterone helps maintain the endometrium, so the ordinary menstrual shedding does not occur. The developing placenta later takes on a larger role in supplying these hormones; the detailed transition belongs to Chapter 15.

Follow the complete chain: trophoblast releases hCG; hCG supports the corpus luteum; the corpus luteum supplies progesterone and estrogen; those hormones support the uterine lining. hCG and progesterone are different signals with different sources and immediate targets. A pregnancy test can detect hCG, but neither a missed period nor a hormone sketch is a personal diagnosis.

External hormones change feedback rather than blocking a duct

Hormonal contraception uses hormones or hormone-like substances to alter reproductive processes. Combined methods supply an estrogen and a progestin, a substance with progesterone-like activity. These signals act through feedback on the hypothalamus and anterior pituitary, reducing the normal FSH/LH stimulation of the ovaries. In particular, suppression of the LH surge helps prevent ovulation. Reduced follicular development is another part of the mechanism.

Read this as supplied signal, changed pituitary output, then changed ovarian activity. It does not mean that all FSH and LH disappear or that every oocyte is destroyed. The teacher slide's wording about "high doses" and being "without FSH and LH" is too absolute for this mechanism. Effects on cervical mucus also make sperm passage more difficult. Hormonal methods differ; this combined-method example does not describe every method in the same way.

Notice what has not been changed: altering hormone feedback does not create a physical barrier against infectious organisms. A combined contraceptive pill does not protect against sexually transmitted infections, or STIs. Preventing conception and reducing infection transmission are different biological functions. This is a comparison of mechanisms, not advice about choosing, dosing or changing a method.

The same feedback reasoning helps with the optional testosterone questions. An external testosterone-like signal can inhibit hypothalamic and pituitary stimulation, reducing LH support for the testes' own testosterone production and disrupting support for sperm production. Effects at body tissues and effects on the control system can therefore differ. Human growth hormone, hGH, is a protein hormone, not an anabolic steroid; the example on teacher slide19 misclassifies it.

Worked example

Compare a source change with a target change

Two constructed models have functioning hypothalamic and pituitary feedback. In Model A, ovarian hormone output falls because follicles respond poorly. In Model B, ovarian tissue is responsive, but an external estrogen/progestin signal produces inhibitory feedback. Predict the direction of pituitary stimulation and explain why both models can disrupt ovulation. Assume no other changes.

- Locate the change in Model A: the ovary produces less inhibitory hormone signal. It is not a failure of the pituitary to send a message.
- Trace A upstream: weaker inhibition permits FSH/LH stimulation to rise. Trace back to the target: poor follicular responsiveness means more stimulation need not restore maturation or ovulation.
- Locate the change in Model B: an external inhibitory signal is supplied. Trace upstream output: normal FSH/LH stimulation and the LH surge can be suppressed. Responsive ovaries still need the appropriate stimulation to complete the normal sequence.
- Compare the results: disrupted ovulation can accompany increased stimulation in A or reduced stimulation in B. The mechanism, not just the final outcome, determines the hormone prediction.
- Check the limit: these deliberately specified models demonstrate feedback. A single measurement from a person would not establish which model, if either, applies.

Optional application

Optional model challenge: hCG is present in early pregnancy, but in a constructed model the corpus luteum cannot respond to it. No other tissue supplies enough progesterone or estrogen to replace that lost output. Would the presence of hCG alone guarantee maintenance of the endometrium? Trace the missing link.

Answer withheld for first attempt

Practise with support

Apply the explanation

Try both questions. Use the hint to revise your thinking after an incorrect answer.

Which signal helps maintain the corpus luteum in early pregnancy?

- The loss of all pituitary activity
- hCG
- A new monthly LH surge

Why does a combined contraceptive pill not protect against an STI?

- FSH is itself a pathogen
- Every STI can enter only during ovulation
- Changing hormonal feedback does not create a barrier to pathogen transmission

Stop and think

Try this before opening the explanation

Would low progesterone in one sample prove that ovulation never occurred?

Answer withheld for first attempt

Optional video support

Hormonal contraception and feedback

As you watch: Connect an external hormone signal to altered pituitary stimulation.

Optional video. Internet access is required. The explanation below covers the idea without the video.

Read the explanation

Combined hormonal methods provide estrogen and a progestin. Their feedback effects can limit FSH and LH stimulation and help prevent ovulation. They do not protect against STIs.

Open required check · When hormone patterns change

Explain what you have learned

This check counts toward chapter progress. Correct the 2 selections to unlock 2 written explanations. Saving writing records your response; it does not grade its accuracy.

Selection 1

In a functioning control system, reduced ovarian hormone feedback can produce which response?

- Increased FSH and LH stimulation
- Movement of the ovaries into the uterus
- A guaranteed new pregnancy
- A complete loss of all pituitary hormone production

Selection 2

Which statement correctly explains one effect of combined hormonal contraception?

- It permanently removes all oocytes
- Feedback can reduce the normal LH surge and prevent ovulation
- It blocks the vas deferens
- It guarantees protection against bacterial and viral STIs

Answer the selections correctly to unlock the writing.

Written explanation 1

Explain why low ovarian hormone output and high pituitary hormone output can occur together.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Written explanation 2

Contrast a cycle without pregnancy with early pregnancy in terms of corpus-luteum activity and endometrial support.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Optional application 1

This is optional written practice. It does not change required progress.

Explain how the hormones in a combined contraceptive pill can reduce the chance of pregnancy. Trace the feedback effect to ovarian activity.

[Write your own response separately.]

Optional response. Select Save response to collect it. Maximum 5000 characters.

← Hormone changes across the menstrual cycle / Next: STIs and reproductive function →